BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022 • Paper Code: 6465

Retyped faithfully from the original board paper — same questions, no changes.

Biology - Paper: I - Objective (ii)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6465

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. Surfactant is mainly composed of:

(A) Lipids
(B) Proteins
(C) Lipoproteins
(D) Carbohydrates

Q2. Appendix arise from blind end of:

(A) Ileum
(B) Caecum
(C) Colon
(D) Rectum

Q3. In which of the given parts of body lactic acid fermentation takes place?

(A) Brain
(B) Muscle
(C) Heart
(D) Liver

Q4. The product of cyclic phosphorylation is/are:

(A) ATP
(B) NADP
(C) NADP and ATP
(D) NADP, ATP and O_2

Q5. Which one of the given is intermediate host for Taenia?

(A) Snail
(B) Pig
(C) Sparrow
(D) Man

Q6. During development of an animal mesoderm layer gives rise to:

(A) Nervous system
(B) Digestive system
(C) Integumentary system
(D) Reproductive system

Q7. The number of ascospores in each ascus are:

(A) 2
(B) 4
(C) 6
(D) 8

Q8. Vascular system is absent in:

(A) Bryophytes
(B) Pteridophytes
(C) Gymnosperms
(D) Angiosperms

Q9. Which one of the given phylums has no flagella?

(A) Euglenophyta
(B) Pyrrophyta
(C) Rhodophyta
(D) Chlorophyta

Q10. Antibiotics are produced by fungi and certain bacteria of groups:

(A) Oomycetes
(B) Basidiomycetes
(C) Ascomycetes
(D) Actinomycetes

Q11. The mysterious brain infection in man is caused by:

(A) Virion
(B) Fungi
(C) Bacteria
(D) Prions

Q12. Nuclear membrane is continuous with:

(A) Endoplasmic reticulum
(B) Golgi bodies
(C) Lysosomes
(D) Peroxisomes

Q13. The optimum pH for pepsin is:

(A) 2.00
(B) 4.50
(C) 5.50
(D) 7.60

Q14. Identify the unsaturated fatty acid.

(A) Acetic acid
(B) Butyric acid
(C) Palmitic acid
(D) Oleic acid

Q15. A method in which pests are destroyed by using some living organism is called:

(A) Biological control
(B) Insecticide control
(C) Cultural control
(D) Pesticide control

Q16. Hydathodes are linked with one of the given processes.

(A) Imbibition
(B) Bleeding
(C) Guttation
(D) Transpiration

Q17. Which one of the given parts of fish body has oxygenated blood?

(A) Sinus venosus
(B) Dorsal aorta
(C) Ventral aorta
(D) Atrium

Biology - Paper: I - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note:- Section B is compulsory. Attempt any 3 questions from Section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8x2 = 16)

i. Why proper arrangement of amino acids is necessary for proteins? Give an example.

ii. What will happen to enzymatic reactions if the temperature becomes 50°C?

iii. Why some enzymes are not produced in active form? Give an example.

iv. Enzymes become denatured in what ways? Explain briefly.

v. What are septate and non-septate hyphae?

vi. Write a brief note on yeast.

vii. Give four parasitic adaptations in Platyhelminthes.

viii. Define Moulting.

ix. What do you know about locusts?

x. Give two differences between osteichthyes and chondrichthyes.

xi. What is compensation point?

xii. How absorption spectrum differs from action spectrum?

Q3.. (Marks 16) Write short answers to any Eight parts. (8x2 = 16)

i. Differentiate between biopesticides and biological control.

ii. How a biologist can help to reduce environmental pollution?

iii. Why thylakoid has grana and intergrana?

iv. Describe the fact that centrifugation is necessary for cellular fractionation.

v. How are limestone deposits formed?

vi. Write two similar and two different characters between algae and green plants.

vii. Give two examples each of red algae and green algae.

viii. What is the name of pornycetes which played infamous role in human history? Comment why it is so notorious?

ix. Which plant group is called arthrophytes and why?

x. Differentiate between monocots and dicots.

xi. What will be direction of flow of blood in the heart of fishes?

xii. Define immunity. What is difference between active and passive immunity?

Q4.. (Marks 12) Write short answers to any Six parts. (6x2 = 12)

i. Define Retroviruses.

ii. Write down few words on the capsule of bacteria.

iii. Write down the names of any two salivary glands. Compare them.

iv. Define Heart Burn.

v. Define Dyspepsia.

vi. Write down the lung capacity of humans.

vii. By listening 'Smoker's Cough' which disease come in our mind? Elaborate.

viii. How muscles get their oxygen?

ix. Has pH any effect on the blood when oxygen combines with haemoglobin?

> Note: SECTION-C (EACH QUESTION CARRIES EIGHT (8) MARKS)

Q5.. (Marks 8) (a) Why is understanding of biodiversity important and how biology is helping to mankind to conserve environment? 4 (b) What is ascent of sap? Explain the Cohesion-Tension Theory? 4

Q6.. (Marks 8) (a) Describe the structure of amino acids and importance of proteins. 2+2 (b) Describe nutrition in Fungi and explain mycorrhizal association. 2+2

Q7.. (Marks 8) (a) Wise use of antibiotics is necessary. What happens if we misuse them? Introduce antibiotics as well. 4 (b) Comprehend various processes that occurred in vascular plants to evolve microphylls and megaphylls. 4

Q8.. (Marks 8) (a) Describe the structure and function of plasma membrane. 4 (b) Explain the digestion of food in Hydra. 4

Q9.. (Marks 8) (a) Describe the structure and function of plasma membrane. 4 (b) Describe any four methods of animal nutrition. 4